

1) Given,

	windows/h	raking leave/h
Sarah	4	6
James	3	5

Parents want 12 windows washed and 25 leaves raked.

- (a) Sarah $\rightarrow$ 6 leaves $\rightarrow$ 1h
 James $\rightarrow$ 5 leaves $\rightarrow$ 1h

Sarah can rake more leaves in a hour than James. So, Sarah has a comparative advantage in raking leaves. So, she should specialize in leaves.

- (b) Sarah specialize $\rightarrow$ leaves
 James " " $\rightarrow$ windows.

$\therefore$ So to rake 25 leaves, Sarah takes $\Rightarrow$

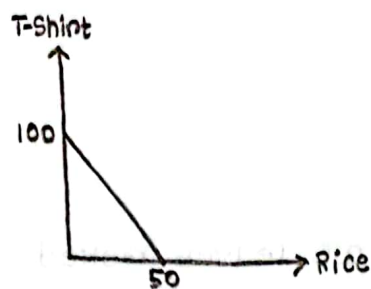
$$\begin{array}{r} 6 \text{ --- } 1\text{h} \\ 25 \text{ --- } \frac{25}{6} \text{ h} \\ \hline = 4.17\text{h} \end{array}$$

$\therefore$ And to rake 12 windows, James takes $\Rightarrow$

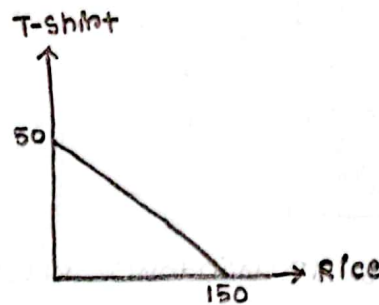
$$\begin{array}{r} 3 \text{ --- } 1\text{h} \\ 12 \text{ --- } \frac{12}{3} \\ \hline = 4\text{h} \end{array}$$

So, together if they work they can complete the work in less than 4.17h, than the usual time.

2) From give graph,



country A



	T-shirt	Rice
Count A	100	50
Count B	50	150

Companitive Advantage: Has lower OC, compared to other one.

%. For Count A, OC of Producing Shirts = $\frac{50}{100}$
 = 0.5 (0.5 Rice sacrifice
 Kore)

For Count B, OC of Producing Shirts = $\frac{150}{50}$
 = 3

%. For Count A, OC of Producing Rice = $\frac{100}{50}$
 on (OC for producing T-shirt) = 2

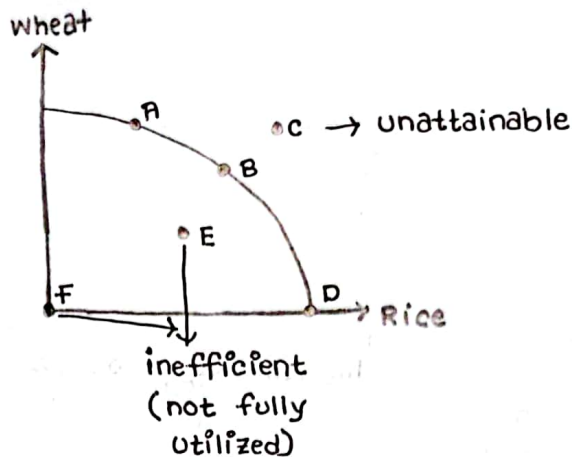
" Count B, " " " " " = $\frac{50}{150}$
 = 0.3

Ans) Country A is not specialized for producing rice since it has to sacrifice 2 shirts in return than country B. B is specialized for producing rice.

CA of Count A → Producing Shirt.

CA of Count B → " " Rice.

3)



(a) A, B and D points efficient since it lies on PPF Curve .

(b) from D $\rightarrow$ B $\rightarrow$ A , OC is increasing.

(c) C point is unattainable since it is outside the curve .

(Does not have enough resources)

(d) F and E are inefficient since they're inside the PPF curve and are not being fully employed .

4) Given,

	Bang	china
Corn	9 acre	8 acre
Rice	3 "	2 "

(a) OC of BD for producing 100 bushels of corn?

(or OC of producing Rice)

$$= \frac{9}{3} \times 100$$

$$= 300$$

(b) OC of China for producing 100 bushels of rice?

(or OC of producing corn)

$$= \frac{2}{8} \times 100$$

$$= 25$$

5) Given,

	olive	Olive oil
Spain	60	11
Greece	8	5
	OC	OC
SP	$\frac{11}{60} = 0.2$	$\frac{60}{11} = 5.5$
Gr	$\frac{5}{8} = 0.6$	$\frac{8}{5} = 1.6$

(a) Which country has CA at Olives?

— Spain (less OC)

(b) Which country has CA at olive oil?

— Greece (less OC)

6) # Alice and Bob math.

(a) Alice $\rightarrow$ 8h Stamina

2h $\rightarrow$ 1 bundle of firewood.

4h $\rightarrow$ 6 coconuts

2h $\rightarrow$ 3 co.

$\therefore$ OC of 1 bundle of wood = $\frac{6}{2} = 3$ coconuts
(for producing coconuts)

Now, OC of 12 coconuts = 4 bundle of firewood.
(for producing woods)

(b) Bob $\rightarrow$ 6h Stamina.

2h $\rightarrow$ 1 bundle wood

1h $\rightarrow$ 1 coconut

$\therefore$ OC for 1 bundle of wood = 2 coconuts.

OC of 6 coconuts in terms of wood = 3 woods $\left(\frac{6}{2}\right) = 3$

(c) CA of Alice $\rightarrow$ Coconut (less OC)

CA of Bob $\rightarrow$ firewood (less OC)

Absolute Advantage $\rightarrow$ Alice

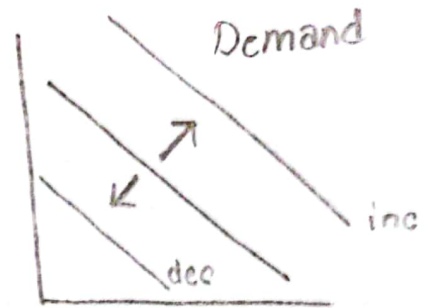
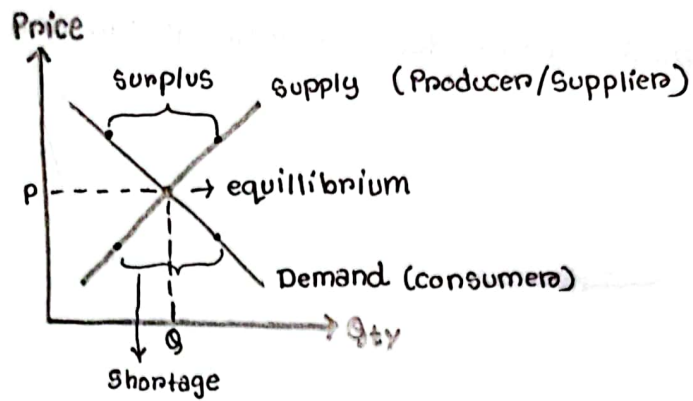
(she can out-gather bob in both if she devotes her time to one or other).

OC of 1 coconut = $\frac{1}{3}$

$$\begin{array}{r} 2 \times 6 \times 12 \\ 2 \times 2 \\ 4 \end{array}$$

$$\begin{array}{r} \frac{1}{2} \times 6 = 3 \\ \left(\frac{2}{1}\right) \end{array}$$

Topic 2: Demand and Supply

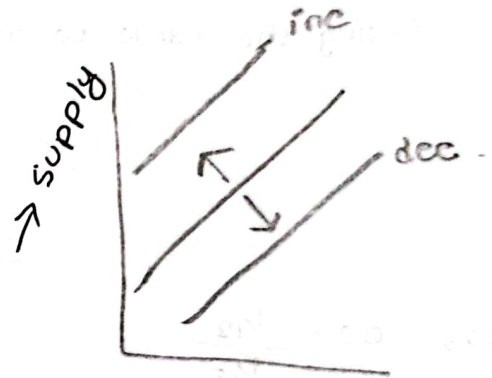


⊞ Law of demand: I. $P \downarrow \rightarrow Q_d \uparrow$
II. $P \uparrow \rightarrow Q_d \downarrow$

↓
Inverse
relation between
Price and Qty.

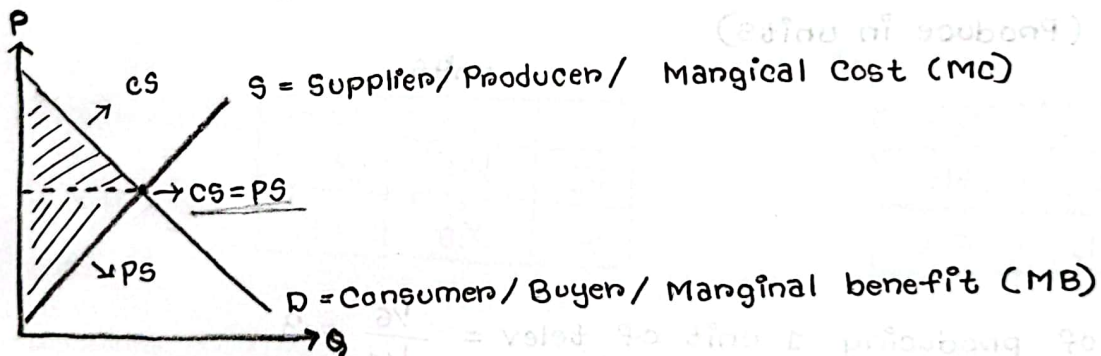
⊞ Law of Supply: I. $P \uparrow \rightarrow Q_s \uparrow$
II. $P \downarrow \rightarrow Q_s \downarrow$

↓
directly
Proportional



• A new topic comes in within it :

• Welfare & Efficiency



% CS = Consumer Surplus. $\Rightarrow$ Just find Area and get value.
PS = Producer Surplus.

7) Given,

It takes each companies to produce a unit of car and a unit of van.

	car	van
Nitol motons	8 days	10 days
Ifad Autos	15 days	12 days

∴ OC of Ifad Autos to produce 2 units of cars.

Taking the table to units.

	car	van
Nitol motons	$\frac{1}{8}$	$\frac{1}{10}$
Ifad Autos	$\frac{1}{15}$	$\frac{1}{12}$

$$\text{So, OC} = \frac{\frac{1}{12}}{\frac{1}{15}}$$

$$= 1.25$$

$$\text{for 2 units} = 1.25 \times 2 \\ = 2.5 \text{ (Ans)}$$

8) Given, (Produce in units)

	Telev	Comp
RL	18h	6h
BL	16h	4h

	Telev	Comp
RL	$\frac{1}{18}$	$\frac{1}{6}$
BL	$\frac{1}{16}$	$\frac{1}{4}$

(a) RL OC of producing 1 unit of telev = $\frac{\frac{1}{6}}{\frac{1}{18}} = 3$

(b) BL OC of producing " " " " = $\frac{\frac{1}{4}}{\frac{1}{16}} = 4$

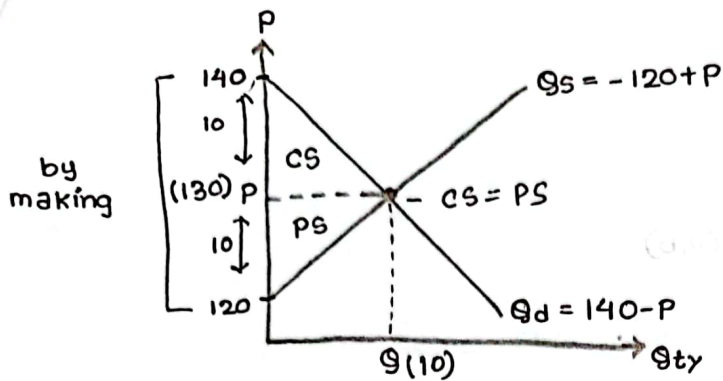
(c) RL OC of producing " " " comp = $\frac{\frac{1}{18}}{\frac{1}{6}} = 0.33$

(d) BL " " " " " " = $\frac{\frac{1}{16}}{\frac{1}{4}} = \frac{1}{4}$

Problem 1

Given, $Q_d = 140 - P$

$Q_s = -120 + P$



(a) Find equilibrium Price.

∴ $Q_d = Q_s$

⇒ $140 - P = -120 + P$

⇒ $2P = 260$

⇒ $P = 130$

(b) Find equilibrium quantity.

∴ $Q_d = 140 - 130$
 $= 10$

(c) Find Consumer surplus.

$= \frac{1}{2} \times 10 \times 10$ (Upper Area)
 $= 50$

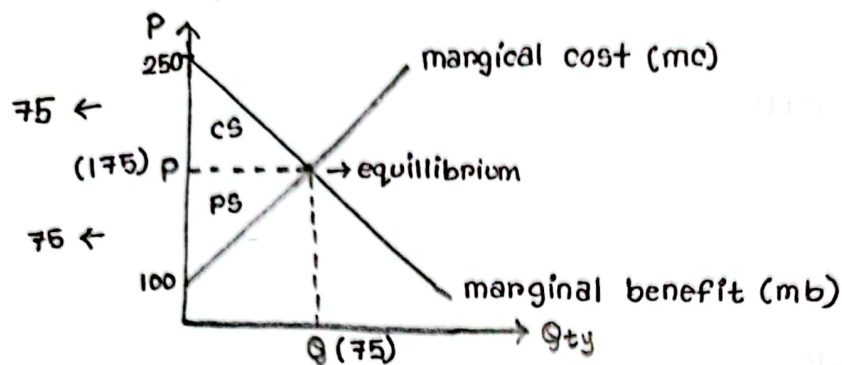
(d) Find Producer Surplus.

$= \frac{1}{2} \times 10 \times 10$ (Lower Area)
 $= 50$

Problem 2

Given, $Q_d = 250 - P$

$$Q_s = -100 + P$$



(a) Equilibrium Price and Quantity.

$$\Rightarrow Q_d = Q_s$$

$$\Rightarrow 250 - P = -100 + P$$

$$\Rightarrow 2P = 350$$

$$\Rightarrow P = 175$$

$$\text{and } Q = 250 - 175 \\ = 75,$$

$$(b) \quad CS = \frac{1}{2} \times 75 \times 75 \\ = 2812.5$$

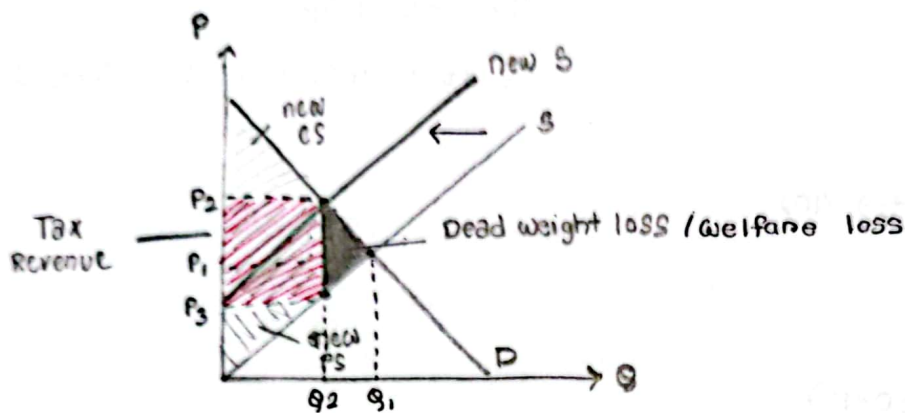
$$(c) \quad PS = \frac{1}{2} \times 75 \times 75 \\ = 2812.5 \quad (\text{Ans})$$

mit key

Problem 3

- Dead-weight loss / effects of tax.

▢ Tax can be imposed on buyers or sellers. Regardless of on whom it is imposed, both the buyers and sellers would be effected.



CS
PS
Total revenue =
 $(P_2 - P_3) \times Q_2$
Total surplus
(after tax)
 $= (CS + PS) + TR$

∴ Total surplus (after tax) = new CS + new PS + tax revenue

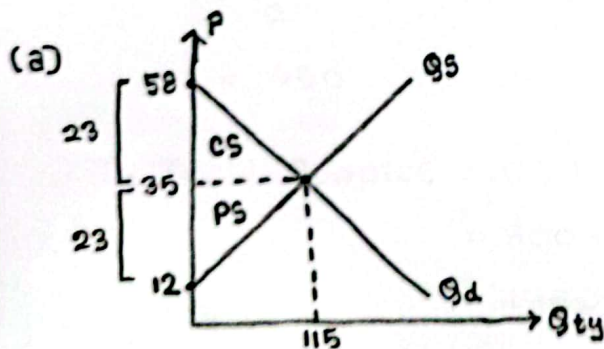
∴ Tax revenue: $(P_2 - P_3) \times Q_2$

∴ Total Surplus (normally) = CS + PS

∴ welfare loss / DWL = $\frac{1}{2} \times (Q_1 - Q_2) \times (P_2 - P_3)$

1) Given, $Q_d = 290 - 5P$

$Q_s = -60 + 5P$



∴ $Q_d = Q_s$

$\Rightarrow 290 - 5P = -60 + 5P$

$\Rightarrow P = 35$

∴ $Q = 290 - (5 \times 35)$
 $= 115$

∴ Total Surplus = $\left(\frac{1}{2} \times 115 \times 23\right) + \left(\frac{1}{2} \times 115 \times 23\right)$
 $= 2645$

$$\frac{50}{5} = 10$$


$$\therefore \text{at } P=40, Q=290-(5 \times 40) \\ = 90$$

2

$$\begin{aligned} -50 + 5P &= 290 - 5P \\ 10P &= 340 \\ P &= 34 \end{aligned}$$

5P-50
290 - (5x34)

$$\begin{aligned} -50 + 5P &= 290 - 5P \\ 5P + 5P &= 290 + 50 \\ P &= 34 \end{aligned}$$

$$\vec{H} =$$

3) Calculate PS, CS (before tax) and CS, PS, dwl (DWL), tax Revenue to the government after tax. (15% tax)

$$P_d = 100 - 2Q$$

$$P_s = 10 + Q$$

∴ Equilibrium point, $P_d = P_s$

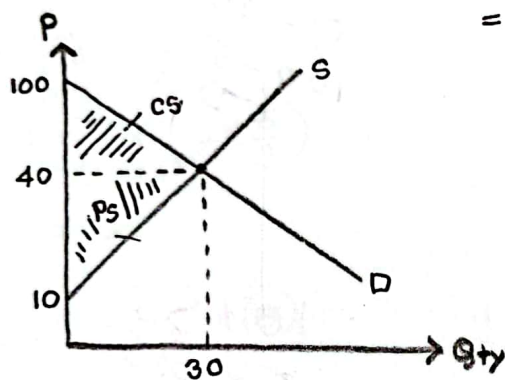
$$\Rightarrow 100 - 2Q = 10 + Q$$

$$\Rightarrow 3Q = 90$$

$$\Rightarrow Q = 30$$

$$\text{and, } P = 100 - (2 \times 30)$$

$$= 40$$



$$\therefore CS = \frac{1}{2} \times 30 \times (100 - 40)$$

$$= 900$$

$$\therefore PS = \frac{1}{2} \times 30 \times (40 - 10)$$

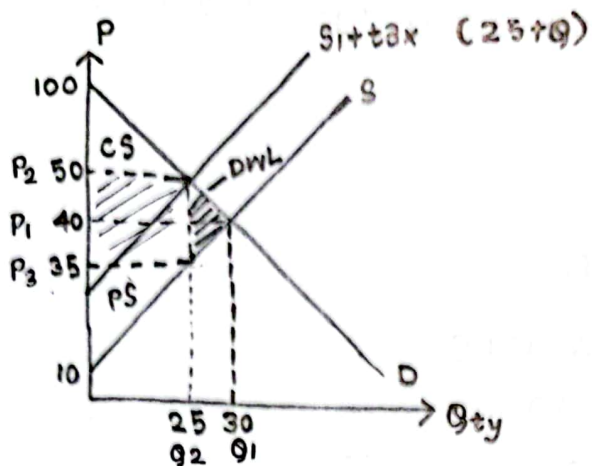
$$= 450$$

$$\therefore \text{Total Surplus} = CS + PS$$

$$= 900 + 450$$

$$= 1350$$

(b) After adding Tax (15%)



$$25 + Q = 100 - 2Q$$

$$3Q = 75$$

$$Q = 25$$

$$P_d = 100 - 2Q$$

$$P_s = 10 + Q$$

$$100 - 2(25) = 100 - 50 = 50$$

$$\therefore CS (\text{after tax}) = \frac{1}{2} \times 25 \times (100 - 50) = 625$$

$$\therefore PS (\text{After tax}) = \frac{1}{2} \times 25 \times (35 - 10) = 312.5$$

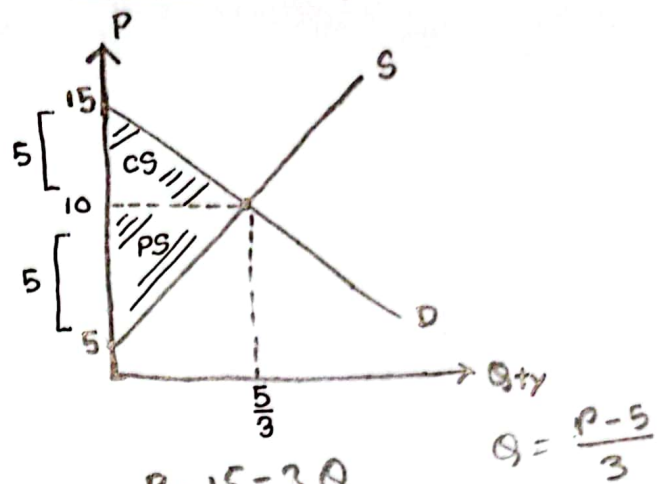
$$\therefore \text{Tax Revenue} = (P_2 - P_3) \times Q_2 = (50 - 35) \times 25 = 375$$

$$\therefore DWL = \frac{1}{2} \times (Q_1 - Q_2) \times (P_2 - P_3) = \frac{1}{2} \times (30 - 25) \times (50 - 35) = 37.5 \text{ (Ans)}$$

$$\text{Total Surplus} = CS + PS + TR = (10 + Q) + (15\% \text{ of } 10 + Q) = 10 + Q + \left(\frac{15}{100} \times 10 \right)$$

Problem

- Given, $P = 5 + 3Q_s$
 $P = 15 - 3Q_d$



(a) Find CS and PS.

∴ Equating both the equations,

$$\Rightarrow 5 + 3Q = 15 - 3Q$$

$$\Rightarrow 6Q = 10$$

$$\Rightarrow Q = \frac{5}{3}$$

$$\therefore P = 5 + \left(3 \times \frac{5}{3}\right)$$

$$= 10$$

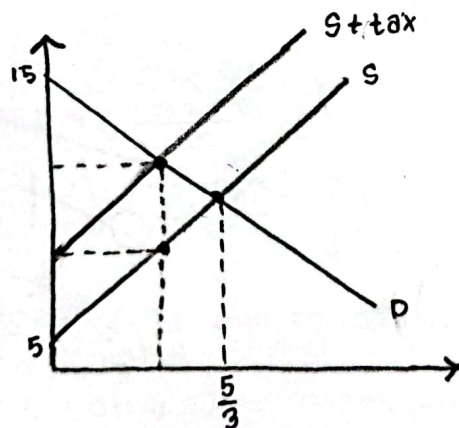
$$I. CS = \frac{1}{2} \times \frac{5}{3} \times (5)$$

$$= 4.167$$

$$II. PS = \frac{1}{2} \times \frac{5}{3} \times 5$$

$$= 4.167$$

(b) Gov adds 10th tax/unit. Find new CS and PS.



$$S + \text{tax} = 5 + 3Q_s + 10$$

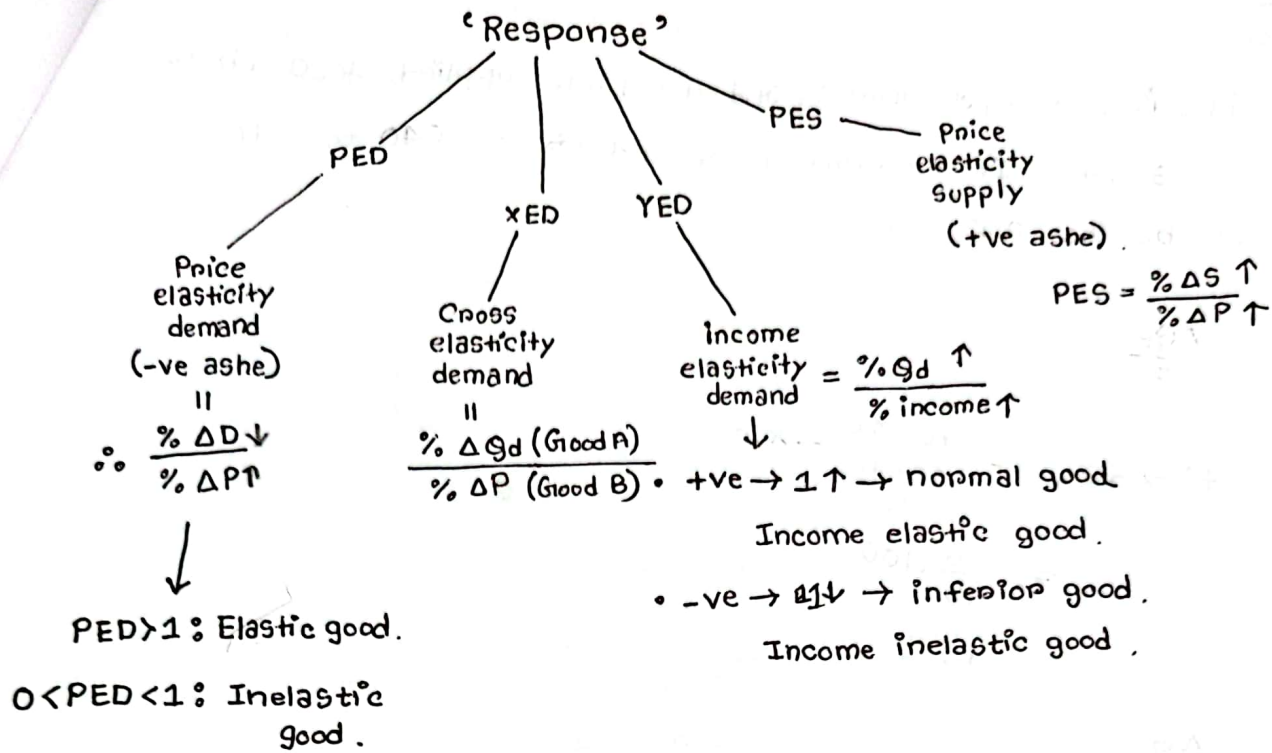
$$= 15 + 3Q_s$$

$$\therefore 15 + 3Q = 15 - 3Q$$

$$Q = 0$$

Topic 4: Elasticity

'Response'



PED = 0, Perfectly inelastic good



PED = ∞, " " elastic good



PED = 1, unit elastic " "

luxury goods

%. mid-point formula, $\% \Delta = \frac{\text{New} - \text{old}}{\frac{(\text{new} + \text{old})}{2}} \times 100$

$$1. \text{ PED} = \left(\frac{\frac{Q_2 - Q_1}{\frac{Q_2 + Q_1}{2}}}{\frac{P_2 - P_1}{\frac{P_2 + P_1}{2}}} \right)$$

• YED → measures ΔQ_d due to 1% change in Income (Y).

%. YED (0 to 1) → necessary goods.

YED > 1 → luxury goods

$$\frac{Q_d \uparrow}{\text{Income} \uparrow}$$

• XED → XED > 0, substitute goods

XED < 0, complementary goods

XED = 0, neutral goods

XED < -1 elastic

XED < 0 inelastic

Practice

- 1) If the PES is 2 for pen drives: and the firm supplied 4000 when the price was £40. If the price increased from £40 to £46, what would be the new Q?

$$\therefore \text{PES} = \frac{\Delta Q_s}{\Delta P}$$

$$\begin{aligned} \therefore \text{Price } 40 \rightarrow 46, & \quad \frac{46-40}{\frac{46+40}{2}} \times 100 \\ & = \frac{6}{43} \times 100 \\ & = 14\% \end{aligned}$$

$$\therefore \text{PES} = \frac{\Delta Q_s}{14}$$

$$\Rightarrow 2 \times 14 = \Delta Q_s$$

$$\Rightarrow \Delta Q_s = 28\%$$

$$\begin{aligned} \text{New } Q &= 4000 + (0.28 \times 4000) \\ &= 5120 \text{ (Ans)} \end{aligned}$$

PES = 2

$$\frac{\Delta Q_s}{\Delta P}$$



- 2) There is a 12% rise in price of tea. As a result, the quantity demanded of milk fall by 34%, and quantity demanded of coffee rise by 14% and Qd of tea fall by 22%.

$$\begin{aligned}\text{Ans) (a) PED of tea} &= \frac{\Delta Q_d}{\Delta P} \\ &= \frac{-22\%}{12\%} \\ &= -1.83\end{aligned}$$

$$\begin{aligned}\text{(b) XED of tea and milk} &= \frac{\Delta Q_d (\text{milk})}{\Delta P (\text{tea})} \\ &= \frac{-34\%}{12\%} \\ &= -2.83\end{aligned}$$

$$\begin{aligned}\text{(c) XED of tea and coffee} &= \frac{\Delta Q_d (\text{coffee})}{\Delta P (\text{tea})} \\ &= \frac{14\%}{12\%} \\ &= 1.17.\end{aligned}$$

Practice

- 1) If the PES is 2 for pen drives and the firm supplied 4000 when the price was £40. If the price increased from £40 to £46, what would be the new Q?

$$\therefore \text{PES} = \frac{\Delta Q_s}{\Delta P}$$

$$\begin{aligned} \therefore \text{Price } 40 \rightarrow 46, & \quad \frac{46-40}{46+40} \times 100 \\ & \quad \frac{6}{92} \times 100 \\ & \quad = 6.52\% \end{aligned}$$

$$\text{PES} = 2$$

$$\frac{\Delta Q_s}{\Delta P}$$

$$\therefore \text{PES} = \frac{\Delta Q_s}{14}$$

$$\Rightarrow 2 \times 14 = \Delta Q_s$$

$$\Rightarrow \Delta Q_s = 28$$

$$\begin{aligned} \text{New } Q &= 4000 + (0.28 \times 4000) \\ &= 5120 \text{ (Ans)} \end{aligned}$$



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